

Symplectic integrators: precession of Mercury's orbit

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1 Introduction

1.1 Background

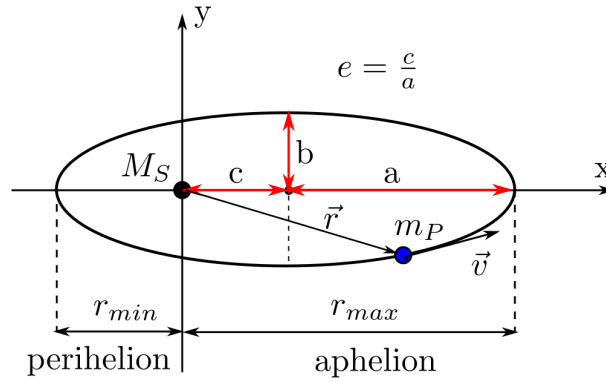


Figure 1: Planet of mass m_P orbits around the star (mass M_S) along the elliptic trajectory, a is semimajor axis, b is semiminor axis, e is eccentricity, $c = \sqrt{a^2 - b^2}$ is linear eccentricity. Source of central force (star) is located at one of two foci of ellipse.

Our planetary system is composed of a massive star (Sun) which is a source of gravitational force for several planets orbiting around it. If two masses are quite distant from each other then their interaction is governed by the force which has simple power inverse ($\sim r^\beta$) form

$$F_g = -\frac{GM_S m}{r^2} \quad (1)$$

According to first Kepler's law any planet must orbit around the Sun on elliptic trajectory and here we silently neglect the perturbative interactions between planets. Example trajectory of a planet is shown in Fig.1. This is a standard picture we have learnt at the basic course on celestial mechanics. Albert Einstein in 1915 had made this picture a little complicated. According to General Relativity theory the space in vicinity of massive object becomes curved what influences a little on the way we measure the distance. For this reason the expression for the classical gravitational force is attached a relativistic correction

$$F_{gr} = -\frac{GM_S m}{r^2} \left(1 + \frac{\alpha}{r^2}\right) \quad (2)$$

where α is the number factor which quantifies the relativistic correction. In solar planetary system all the planets but three orbit on approximately circular trajectories, the other three are Mercury, Mars and Pluto (dwarf planet) and have elliptic orbits defined by large eccentricities [see Fig.1]: 0.206, 0.093 and 0.248, respectively. Among these three, Mercury is the most interesting because the astronomers have observed that its elliptical orbit slowly precesses in its motion around the Sun, that is the elliptic

orbit is not stable but rotates with time. This effect is called *perihelion precession of Mercury*, its magnitude is very small and equals ~ 43 angle seconds/century (1 angle degree = 60 angle minutes = 3600 angle seconds).

Now arise a question: does this precession is an effect of gravitational perturbation due to other planets or it is a relativistic effect? To answer to this question we need to carry on the numerical simulation of the motion of Mercury accounting for the relativistic effect (Eq.2) and neglect interactions with other planets. If we were able to observe in simulation the rotation of perihelion we would determine the angular velocity of precession and then compare it with the astronomical data - this is the goal of this project. In simulations we will use the symplectic integrator for solving ordinary differential equations (ODE), these are known to retain the energy of the Hamiltonian system with high accuracy and the trajectory as well.

1.2 Equations of motion

In the following we consider the system composed of massive non-moving star of mass M_S located at the origin of Cartesian coordinates system $\vec{r}_S = [0, 0]$ and much lighter planet (Mercury) of mass m orbiting around the Sun which is described by position vector $\vec{r} = [x, y]$ and momentum/velocity vector $\vec{p} = [p_x, p_y] = m[v_x, v_y]$. Hamiltonian for this system is following ($U(\vec{r})$ is the gravitational potential energy)

$$H(\vec{r}, \vec{p}) = \frac{\vec{p}^2}{2m} + U(\vec{r}), \quad \vec{F} = -\nabla_{\vec{r}} U(\vec{r}) \quad (3)$$

and the equations of motion (EOMs)

$$\begin{cases} \vec{v} = \frac{d\vec{r}}{dt} = \frac{\partial H}{\partial \vec{p}} = \frac{\vec{p}}{m} \\ \vec{F} = \frac{d\vec{p}}{dt} = -\frac{\partial H}{\partial \vec{r}} = -\frac{GM_S m}{r^2} \left(1 + \frac{\alpha}{r^2}\right) \frac{\vec{r}}{r} \end{cases} \quad (4)$$

The astronomical quantities expressed in SI units have large magnitudes which are not very handy, so we introduce the more natural astronomical units: for length it is the distance of Earth from Sun (1 AU) while for time it is the one Earth year

$$length \rightarrow 1 \text{ AU} = 1.5 \cdot 10^{11} \text{ [m]}, \quad time \rightarrow [\text{year}] \quad (5)$$

With these assumptions we can carry out nondimensionalization of EOMs, to do this we recall the condition under which the elliptic orbit is stable (closed), namely, it is due to the repelling centrifugal force and attractive gravitational one (without relativistic correction) counteract each other

$$\frac{mv^2}{r} = \frac{GM_S m}{r^2} \implies v = 2\pi \left[\frac{1 \text{ AU}}{\text{year}} \right] \implies GM_S = 4\pi^2 \left[\frac{\text{AU}^3}{\text{year}} \right] \quad (6)$$

This procedure provide us with the nondimensionalized expressions for EOMs

$$\begin{cases} \frac{d\vec{r}}{dt} = \vec{v} \\ \frac{d\vec{v}}{dt} = \frac{1}{m} \frac{d\vec{p}}{dt} = \frac{\vec{F}_{gr}}{m} = -\frac{4\pi^2}{r^2} \left(1 + \frac{\alpha}{r^2}\right) \frac{\vec{r}}{r} \end{cases} \quad (7)$$

1.3 Algorithm for symplectic integrator

We solve the EOMs with symplectic integrator of order $p = 4$ with parametrization by Neri

$$a_1 = a_4 = \frac{1}{2(2 - 2^{1/3})}, \quad a_2 = a_3 = \frac{1 - 2^{1/3}}{2(2 - 2^{1/3})} \quad (8)$$

$$b_1 = b_3 = \frac{1}{(2 - 2^{1/3})}, \quad b_2 = \frac{-2^{1/3}}{(2 - 2^{1/3})}, \quad b_4 = 0 \quad (9)$$

Below is given pseudocode implementing this integrator for EOMs defined in Eq.7. Note however that the momentum vector ($\vec{p}$) is replaced with velocity vector ($\vec{v}$), this allows to remove from EOMs the mass of Mercury

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initialization:   $t_{max}$ ,  $n$ ,  $\alpha$ ,  $k_{max}$  = order of method,
                  $x = x(0)$ ,  $y = y(0)$ ,  $v_x = v_x(0)$ ,  $v_y = v_y(0)$ 
                  $\Delta = \frac{t_{max}}{n}$ 

time=0
for i=1 to n do

    symplectic subiteration:
         $x^{old} \leftarrow x$ 
         $y^{old} \leftarrow y$ 
         $v_x^{old} \leftarrow v_x$ 
         $v_y^{old} \leftarrow v_y$ 
        for k=1 to  $k_{max}$ 
             $x^{new} \leftarrow x^{old} + a_k \cdot V_x^{old} \cdot \Delta t$ 
             $y^{new} \leftarrow y^{old} + a_k \cdot V_y^{old} \cdot \Delta t$ 
             $r^{new} \leftarrow \sqrt{(x^{new})^2 + (y^{new})^2}$ 
             $w^{new} \leftarrow -\frac{4\pi^2}{(r^{new})^3} \left(1 + \frac{\alpha}{(r^{new})^2}\right)$ 
             $v_x^{new} \leftarrow v_x^{old} + b_k \cdot w^{new} \cdot x^{new} \cdot \Delta t$ 
             $v_y^{new} \leftarrow v_y^{old} + b_k \cdot w^{new} \cdot y^{new} \cdot \Delta t$ 

            save values for next subiteration:
             $x^{old} \leftarrow x^{new}$ 
             $y^{old} \leftarrow y^{new}$ 
             $v_x^{old} \leftarrow v_x^{new}$ 
             $v_y^{old} \leftarrow v_y^{new}$ 
        end do

        save new values:
         $x \leftarrow x^{new}$ 
         $y \leftarrow y^{new}$ 
         $v_x \leftarrow v_x^{new}$ 
         $v_y \leftarrow v_y^{new}$ 
         $time+ = \Delta t$ 

        send output:  $x, y$ 

end do

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1.4 Parametrization and initial conditions

For the Mercury and Sun we use the following astronomical data

$$a = 0.397098 \text{ AU} \quad (\text{semimajor axis}) \quad (10)$$

$$T_{\text{Mercury}} = 87.9691 \text{ days} = 0.240846 \text{ year} \quad (\text{time period}) \quad (11)$$

$$M_S = 1.998 \cdot 10^{30} \text{ kg} \quad (\text{mass of Sun}) \quad (12)$$

$$m = 2.4 \cdot 10^{23} \text{ kg} \quad (\text{mass of Mercury}) \quad (13)$$

$$e = 0.206 \quad (\text{eccentricity}) \quad (14)$$

while for setting the initial conditions we may resort to the smallest (perihelion) and to the largest (aphelion) distances from the Sun

$$\text{perihelion: } r_{\min} = a(1 - e), \quad v_{\max} = \sqrt{GM_S} \sqrt{\frac{1 + e}{a(1 - e)} \left(1 + \frac{m}{M_S}\right)} \quad (15)$$

$$\text{aphelion: } r_{\max} = a(1 + e), \quad v_{\min} = \sqrt{GM_S} \sqrt{\frac{1 - e}{a(1 + e)} \left(1 + \frac{m}{M_S}\right)} \quad (16)$$

$$(17)$$

where: a - semimajor, e - eccentricity, $GM_S = 4\pi^2$.

For the initial conditions we assume

$$\vec{r}(t = 0) = [r_{\max, 0}] \quad (18)$$

$$\vec{v}(t = 0) = [0, v_{\min}] \quad (19)$$

1.5 Localization of perihelion and aphelion points

Perihelion and aphelion are the smallest and the largest distance from the Sun, respectively [see Fig.1]. So in the course of simulation, denoting $\vec{r}_i = \vec{r}(t_i)$ and $r_i = |\vec{r}_i|$, we can detect both points by checking in each time iteration the set of conditions

$$\text{perihelion: } r_i < r_{i-1} \quad \text{and} \quad r_i < r_{i+1} \quad (20)$$

$$\text{aphelion: } r_i > r_{i-1} \quad \text{and} \quad r_i > r_{i+1} \quad (21)$$

2 Practical part

1. Implement on computer the symplectic algorithm given in Sec.1.3. In all simulations assume initial conditions as for *aphelion* [Sec.1.4] position. Remember that you use nondimensionalized variables (length=[AU], time=[year]).
2. Make the first test simulation for $t_{\max} = 0.95 \cdot T_{\text{Mercury}}$, $\Delta t = 10^{-4}$ and $\alpha = 0$ (relativistic effect is turned off), draw the plot showing the trajectory, it will look as almost closed. In this test you partly confirm that the program works correctly and the results are in accordance with the initial conditions and the time period of Mercury motion around the Sun.
3. Make the second test of stability of Mercury orbit for much longer simulation time $t_{\max} = 100 \cdot T_{\text{Mercury}}$, $\Delta t = 10^{-4}$ and $\alpha = 0$. Consecutive ellipses should overlap perfectly, that means the energy is retained with high accuracy. Make a plot showing this trajectory [see Fig.2 in Sec.3].

4. Make a test of precession, assume $t_{max} = 4 \cdot T_{Mercury}$, $\Delta t = 10^{-4}$ and $\alpha = 0.01$. This value of α is six order of magnitude larger than the real one, but it allows to observe the precession in short time scale and confirm that the relativistic correction introduced in Eq.2 really works. In simulation detect the positions of consecutive perihelions and aphelions by applying the conditions given in Sec.1.5, write these positions to file and then mark these points in trajectory plot [see Fig.2 in Sec.3]. .

5. Checking the relativistic effect.

Although our program works fine, we can not directly simulate the trajectory for real relativistic parameter ($\alpha = 1.1 \cdot 10^{-8} \text{AU}^2$) and detect the precession, because of the smallness of α . In other words any relativistic effect in numerical data would be hidden by numerical errors such as rounding errors and truncation errors during integration of ODE. However, we may solve this problem in other way. The relativistic correction is the result of first order perturbation calculations, it means that any observed effect of this correction should linearly depend on value of α . Keeping this in mind, we may by first, calculate the angular velocities of precession for other, larger values of α , i.e. $\alpha_1, \alpha_2, \dots$, by second, fit the straight line (linear regression) to gathered data, and by third, estimate the velocity of precession for real α just by inserting it to the equation of fitting line. We will follow this route, it requires making the following steps

- a) Assume initial conditions as before and set the parameters $\Delta t = 10^{-5}$ (we need the finer time-space resolution) and $t_{max} = 3$. Make short simulations for: $\alpha_j = \alpha_{max}/2^j$ for $\alpha_{max} = 0.001$ and $j = 0, 1, 2, 3, 4, 5, 6$. In each simulation, for particular value of α_j , detect two consecutive perihelion points, that is, the angles θ_I and θ_{II} as well as corresponding times t_I and t_{II} . The for given position vector $\vec{r}$ the angle can be calculated with standard library function (C/C++)

$$\theta = \text{atan2}(y, x) \quad (22)$$

having at hand these factors the angular velocity of precession is easy to compute

$$\omega(\alpha_j) = \frac{d\theta}{dt} \approx \frac{\theta_{II} - \theta_I}{t_{II} - t_I} \quad (23)$$

Write all pairs (α_j, ω_j) to a file.

- b) Make a linear fit $\omega = a \cdot \alpha$ for data saved in file, remember there is no constant shift in this relation because relativistic effects vanish for $\alpha = 0$. Here you can exploit the GNUPLOT functionality using just two commands (first open the Gnuplot internal terminal then write)

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f (x)=a*x
fit f(x) 'filename' via a
```

program will return/display the value of slope factor a .

- c) Draw the plot for pairs of gatered data (α_j, ω_j) for $j = 0, 1, \dots, 6$, add the fitting line for comparison [see Fig.3 in Sec.3].
- d) Insert the real value $\alpha = 1.1 \cdot 10^{-8}$ to fitting expression $\omega = a \cdot \alpha$ and calculate the precession velocity. Compare your result with the real one $\alpha = 42.9799 \text{ arcsec/century}$.

6. At home prepare the report including short theoretical background for symplectic integrator, precession of Mercury and results with comments. Answer if the obtained value of angular velocity of Mercury precession does confirm the theoretical relativistic prediction of curving the space-time in vicinity of massive object? Try to explain the difference between your result and the real value, what can be the source of this discrepancy? Can you calculate the precession using the real value of α , what you get as the outcome of simulation?

3 Example results

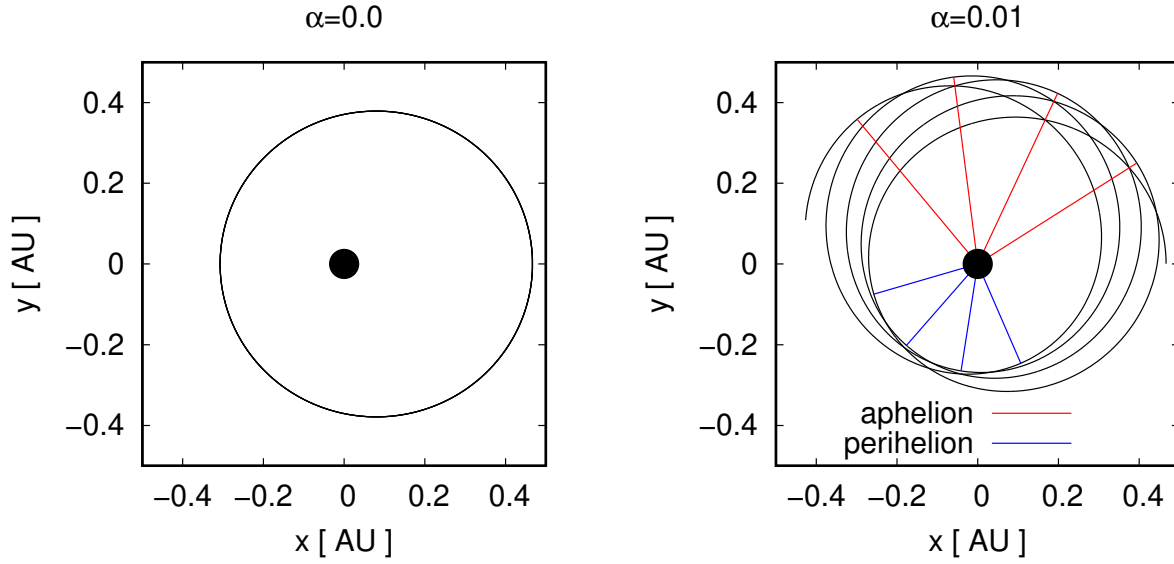


Figure 2: (LEFT) Stable orbit of Mercury without relativistic correction ($\alpha = 0$), and, (RIGHT) precession of perihelion of Mercury for extremely large deviation of central force from inverse power law $1/r^2$ ($\alpha = 0.01$).

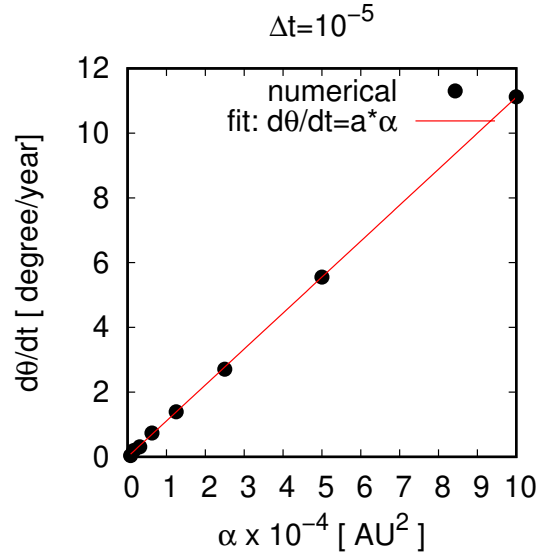


Figure 3: Numerical data of precession velocity as function of α_j (dots) and the fit $\omega = a \cdot \alpha$ (red curve).